

Chapter 1: Integers

10 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-INT-C-01 compares, orders and calculates with integers to solve problems.

NAME _____

CLASS _____

CHAPTER

1

1.01 Integers and opposites

1.02 The number line

1.03 Comparing and ordering integers

1.04 Adding integers

1.05 Subtracting integers

1.06 Multiplying integers

1.07 Dividing integers

1.08 Order of operations with integers

1.09 Integers on a calculator

1.10 Integer problems

How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.01	Integers and opposites	3		☐
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1.04	Adding integers	15		☐
1.05	Subtracting integers	19		☐
1.06	Multiplying integers	23		☐
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1.08	Order of operations with integers	31		☐
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In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers

About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
1.01

Integers and opposites

Date _____

Learning intentions

We are learning to:

- understand what an integer is
- write the opposite of an integer

Success criteria

I can:

- ☐ describe a real situation with a positive or negative number
- ☐ write the **opposite** of any integer
- ☐ explain why 0 is its own opposite

Key terms Write the definition as your teacher explains it.

Integer

Negative number

Opposite

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write an integer for each situation, or its opposite.

- a** Write an integer for: a temperature 6°C below zero; a gain of 4 metres.

Your teacher works this out. Copy every step.

- b** Write the opposite of: -9 , 15 , 0

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write an integer for each situation.

12 QUESTIONS

REMEMBER **INTEGERS** are whole numbers: $\dots, -2, -1, 0, 1, 2, \dots$

- 1** 5°C below zero

- 2** a profit of \$40

- 3** 12 m below sea level

- 4** a rise of 8 floors

- 5** a loss of 3 kg

- 6** 200 m above sea level

Questions 7–12: Write the opposite of each integer.

- 7** -8

- 8** 12

- 9** -25

- 10** 0

- 11** 40

- 12** -1

YOU DO

On your own: **Easy** Set B. Write the opposite of each integer.

6 QUESTIONS

1 -2

4 -10

2 -23

5 8

3 56

6 0



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use integers to describe the movement.

- a** A lift goes 3 floors down, then 3 floors up. Write both moves as integers. Where does it end?

Your teacher works this out. Copy every step.

- b** Which is further from 0: -8 or 6? Explain using opposites.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Opposites in real life.

6 QUESTIONS

REMEMBER Negative numbers are below zero.

- 1 What is the opposite of "3 steps forward"? Write both as integers.

- 4 Which is further from 0: -12 or 9?

- 2 A diver is at -14 m. What is the opposite position?

- 5 Write two integers that are 5 units from 0.

- 3 Write the opposite of the opposite of -7.

- 6 A bank balance of -\$60 means what?

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use opposites to find the number.

- a** The opposite of a number is 25. What is the number?

Your teacher works this out. Copy every step.

- b** List the integers whose distance from 0 is exactly 7.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER The **OPPOSITE** is the same distance from 0 on the other side.

- 1** Is -0 different from 0? Explain.

- 3** List all integers whose distance from 0 is less than 3.

- 2** The opposite of a number is 18. What is the number?

- 4** Does every integer have an opposite? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write the opposite of -15 .

Level 2

Write an integer for "a drop of 9°C ", then write its opposite.

Level 3

The opposite of a number is -31 . What is the number? Explain.

 Extension · Stretch yourself Worked example

The opposite of n is written $-n$. If $n = -4$, what is $-n$?

1. $-n$ means “the opposite of n ”
2. The opposite of -4 is 4

Answer: $-n = 4$

YOU DO

On your own: read the worked example, then try these.

- E1 If $n = -11$, find $-n$ and $-(-n)$.

- E2 Is $-n$ always a negative number? Give an example.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Integers** are whole numbers: ..., -2 , -1 , 0 , 1 , 2 , ...
 2. Negative numbers are below zero.
 3. The **opposite** is the same distance from 0 on the other side.
 4. 0 is its own opposite.

 Level 1

Write -7 as a real situation.

1. Negative means below or less

Answer: 7°C below zero

 Level 2

Opposite of -23 ?

1. Same distance, other side of 0

Answer: 23

 Level 3

Which integers are 4 from 0?

1. One on each side of 0

Answer: 4 and -4

 Extension

If $n = -6$, find $-n$.

1. Opposite of -6

Answer: 6

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LESSON
1.02

The number line

Date _____

Learning intentions

We are learning to:

- show integers on a number line
- find integers between two numbers

Success criteria

I can:

- ☐ draw a number line with an even scale
- ☐ mark positive and negative integers in the right places
- ☐ list the integers **between** two numbers

Key terms Write the definition as your teacher explains it.

Number line

Positive

Negative

Notes

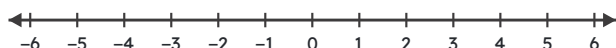
Level 1 · Getting started

Easy · 8 questions

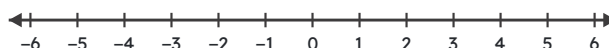
WE DO With your teacher: copy what they write on the board.

Example 1 Mark the integers on the number line.

a Mark 3, -2, 0 and -5 on the number line.



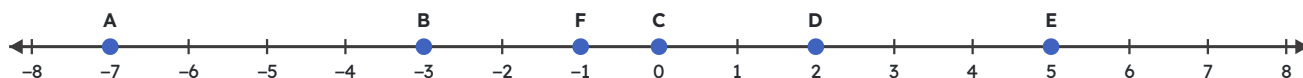
b Mark the integers between -3 and 2.



YOU DO On your own: **Easy** Set A. Which integer is shown by each letter?

6 QUESTIONS

REMEMBER Numbers **INCREASE** to the right.



1 A

4 D

2 B

5 E

3 C

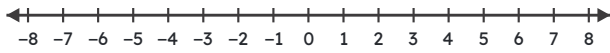
6 F

YOU DO

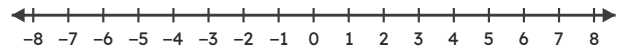
On your own: **Easy** Set B. Mark each set of numbers on the number line.

2 QUESTIONS

- 1 6, -3, 5, 1, -1



- 2 -7, 4, -4, 0



Level 2 · Building up

Medium · 4 questions

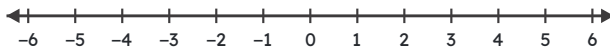
WE DO

With your teacher: copy what they write on the board.

Example 2

Mark the integers, then answer the question.

- a Mark -4, 2, -1 and 5.



- b Which integer is 3 to the left of -2?

Your teacher works this out. Copy every step.

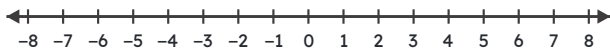
YOU DO

On your own: **Medium** Set C. Mark each set on the number line.

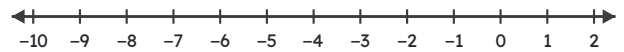
4 QUESTIONS

REMEMBER Negatives are left of 0.

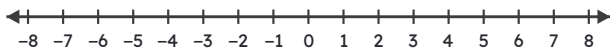
- 1 integers between -2 and 5



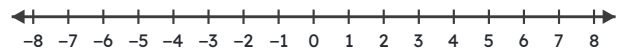
- 3 integers between -8 and -3



- 2 integers less than 1 and greater than -6



- 4 all even integers from -6 to 6



Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Choose a scale, then mark the integers.

a Mark -30 , 10 and -15 using steps of 5 .



b How many integers are between -6 and 4 ?

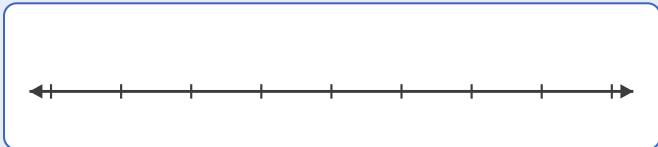
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Number lines with a scale.

4 QUESTIONS

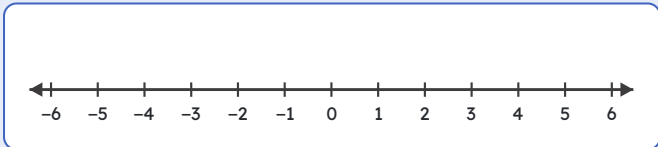
REMEMBER Use an **EVEN SCALE**: equal gaps.

1 Mark -20 , 15 , -5 and 30 . Use steps of 10 .



3 How many integers are between -9 and 9 ?

2 Mark the integers more than -4 but at most 2 .



4 Which integer is halfway between -10 and 4 ?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

List the integers between -3 and 1 .

Level 2

Which is further left on a number line: -8 or -2 ?

Level 3

Which integer is halfway between -9 and 3 ?

Extension · Stretch yourself

Worked example

How many integers are between -15 and 12 ?

1. From -14 to -1 : 14 integers
2. Then 0: 1 integer
3. From 1 to 11: 11 integers

Answer: $14 + 1 + 11 = 26$ integers

YOU DO

On your own: read the worked example, then try these.

E1 How many integers are between -30 and 25 ?

E2 Which integer is halfway between -17 and 5 ?

Lesson summary

Missed the lesson? Start here.

Remember

1. Numbers **increase** to the right.
3. Use an **even scale**: equal gaps.

2. Negatives are left of 0.
4. "Between" does not include the end numbers.

Level 1

Where is -3 ?

1. 3 steps left of 0

Answer: 3 marks left of zero

Level 2

Integers between -2 and 2 ?

1. Do not include -2 and 2

Answer: $-1, 0, 1$

Level 3

Mark -20 on a scale of 10s

1. Each gap is 10
2. 2 gaps left of 0

Answer: second mark left of 0

Extension

Halfway between -6 and 2 ?

1. Distance 8, half is 4
2. $-6 + 4$

Answer: -2

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LESSON
1.03

Comparing and ordering integers

Date _____

Learning intentions

We are learning to:

- compare integers using $<$ and $>$
- put integers in ascending and descending order

Success criteria

I can:

- ☐ use a number line to decide which integer is larger
- ☐ write true statements with $<$ and $>$
- ☐ order a set of integers from smallest to largest and largest to smallest

Key terms Write the definition as your teacher explains it.

Ascending

Descending

$<$ and $>$

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Decide true or false, then order the integers.

a True or false: $-4 > 2$? $-2 < -3$?

b Write in ascending order: 3, -6, 0, -1, 5

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write true (T) or false (F).

14 QUESTIONS

REMEMBER Further **RIGHT** on the number line = **LARGER**.

1 $-4 > 2$

2 $-2 < -3$

3 $2 > -1$

4 $-11 < -10$

5 $-3 > -4$

6 $-9 > -2$

Questions 7–14: Write $<$ or $>$ in the box.

7 $-5 \square -1$

8 $-9 \square -3$

9 $0 \square -4$

10 $-6 \square -7$

11 $3 \square -3$

12 $-10 \square 2$

13 $-15 \square -20$

14 $-30 \square -3$

YOU DO

On your own: **Easy** Set B. Write $<$ or $>$ between each pair.

6 QUESTIONS

1 $5 \square -5$

4 $-12 \square -21$

2 $-7 \square -1$

5 $-1 \square 1$

3 $0 \square -3$

6 $-100 \square -99$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write $<$ or $>$, then order the integers.

a Write $<$ or $>$: $-6 \square -9$

Your teacher works this out. Copy every step.

b Descending: $-2, 8, -15, 0, 3$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Order each set.

6 QUESTIONS

REMEMBER $<$ means “is less than”; $>$ means “is greater than”.

1 Ascending: $4, -2, 7, -8, 0$

4 Descending: $-7, -2, -11, 6, -1$

2 Descending: $-9, 14, 3, 0, -5, -10, -1$

5 Ascending: temperatures $-4^{\circ}\text{C}, 3^{\circ}\text{C}, -12^{\circ}\text{C}, 0^{\circ}\text{C}, -1^{\circ}\text{C}$

3 Ascending: $-15, 12, -3, -20, 8$

6 Descending: heights $40\text{ m}, -8\text{ m}, -35\text{ m}, 12\text{ m}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Order the integers.

a Ascending: $-(-3)$, -5 , 1 , $-(-2)$

Your teacher works this out. Copy every step.

b Write 3 integers between -12 and -8 in ascending order.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Challenge.

4 QUESTIONS

REMEMBER **ASCENDING**: smallest to largest.

1 Write 3 integers that are less than -5 but greater than -10 .

3 Put in ascending order: -1 , -10 , -100 , -1000 , 0

2 Which is the smallest: $-(-6)$, -8 or $-(-10)$?

4 Is it true that the larger the number after a negative sign, the smaller the integer? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

True or false: $-7 > -3$?

Level 2

Write in descending order: 2 , -5 , 0 , -1

Level 3

Write in ascending order: -12 , 5 , $-(-8)$, -20 , 0

 Extension · Stretch yourself Worked example

Write in ascending order: -3 , 2 , $-(-5)$, -7 , 0

- $-(-5)$ is the opposite of -5 , which is 5
- Now order: -7 , -3 , 0 , 2 , 5

Answer: -7 , -3 , 0 , 2 , $-(-5)$

YOU DO

On your own: read the worked example, then try these.

- E1 Order from smallest: $-(-2)$, -4 , $-(-4)$, 3 , $-(-9)$

- E2 Find two integers between $-(-3)$ and $-(4)$.

Lesson summary

Missed the lesson? Start here.

Remember 1. Further **right** on the number line = **larger**.

2. $<$ means “is less than”; $>$ means “is greater than”.

3. **Ascending**: smallest to largest.

4. **Descending**: largest to smallest.

 Level 1

Is $-5 < -1$?

- -5 is further left

Answer: True

 Level 2

Order -2 , 4 , -8 , 1

- Smallest is the most negative

Answer: -8 , -2 , 1 , 4

 Level 3

Descending: -3 , -30 , 3 , 0

- Largest first

Answer: 3 , 0 , -3 , -30

 Extension

Order $-(-4)$, -6 , 2

- $-(-4) = 4$

Answer: -6 , 2 , 4

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.04

Adding integers

Date _____

Learning intentions

We are learning to:

- add positive and negative integers

Success criteria

I can:

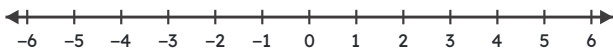
- ☐ use a number line to add integers
- ☐ know that adding a negative is the same as subtracting
- ☐ add three or more integers

Key terms Write the definition as your teacher explains it.**Sum****Adding a negative**

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate. Use the number line for part a.**a** $3 + (-4)$. Use the number line.**b** $-8 + (-2)$ and $-4 + 8$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each sum.

16 QUESTIONS

REMEMBER Adding a **POSITIVE**: move right.

1 $5 + (-2)$

2 $-3 + 7$

3 $-6 + (-1)$

4 $10 + (-13)$

5 $-5 + (-5)$

6 $-3 + 6$

Questions 7–16: evaluate each sum.

7 $-6 + 9$

8 $-8 + (-5)$

9 $3 + (-7)$

10 $-4 + 10$

11 $-12 + 5$

12 $6 + (-11)$

13 $-3 + (-6)$

14 $15 + (-8)$

15 $-20 + 4$

16 $2 + (-9)$

YOU DO

On your own: **Easy** Set B. Evaluate each sum.

6 QUESTIONS

1 $-9 + 4$

4 $-1 + 15$

2 $8 + (-8)$

5 $20 + (-25)$

3 $-12 + (-3)$

6 $-7 + (-7)$

Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $-5 + (-4) + (-2)$

Your teacher works this out. Copy every step.

b $-8 + 5 + 4$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Three or more integers.

6 QUESTIONS

REMEMBER Adding a **NEGATIVE**: move left.

1 $-5 + (-4) + (-2)$

4 $3 + (-9) + 10$

2 $-8 + 5 + 4$

5 $-20 + 15 + (-6)$

3 $-7 + 6 + (-7)$

6 $-1 + (-2) + (-3) + (-4)$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** It is -7°C and the temperature rises 12°C . New temperature?

Your teacher works this out. Copy every step.

- b** Find the missing number: $\square + 9 = -4$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Adding integers in context.

4 QUESTIONS

REMEMBER Same signs: add and keep the sign.

- 1** The temperature is -6°C and rises 11°C . What is the new temperature?

- 3** Maya has \$25 and spends \$40 on credit. What is her balance?

- 2** A lift at level -2 goes up 7 floors. Which level is it at?

- 4** Find the missing number: $-9 + \square = -3$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-6 + 9$

Level 2

$-4 + (-8) + 5$

Level 3

It is -3°C and the temperature drops 5°C , then rises 9°C . What is the temperature now?

 Extension · Stretch yourself Worked exampleFind the missing number: $\square + (-14) = -5$

1. Think: what plus -14 gives -5 ?
2. Undo by adding 14: $-5 + 14 = 9$
3. Check: $9 + (-14) = -5$ ✓

Answer: 9

YOU DO

On your own: read the worked example, then try these.

E1 $\square + (-8) = -20$

E2 $-13 + \square + 6 = 0$

Lesson summary

Missed the lesson? Start here.

Remember

1. Adding a **positive**: move right.
2. Adding a **negative**: move left.
3. Same signs: add and keep the sign.
4. Different signs: subtract, keep the sign of the bigger number.

 Level 1

$4 + (-6)$

1. Start at 4, move 6 left

Answer: -2

 Level 2

$-5 + (-3)$

1. Same signs: $5 + 3 = 8$, keep -

Answer: -8

 Level 3

$-9 + 4 + (-2)$

1. $-9 + 4 = -5$
2. $-5 + (-2) = -7$

Answer: -7

 Extension

$\square + (-4) = 1$

1. $1 + 4 = 5$

Answer: 5

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LESSON
1.05

Subtracting integers

Date _____

Learning intentions

We are learning to:

- subtract positive and negative integers

Success criteria

I can:

- ☐ subtract a positive integer by moving left
- ☐ know that subtracting a negative is the same as adding
- ☐ simplify expressions with + and – signs

Key terms Write the definition as your teacher explains it.**Difference****Subtracting a negative**

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate.**a** $7 - 11$ and $-4 - 8$

Your teacher works this out. Copy every step.

b $3 - (-4)$ and $-5 - (-9)$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each expression.

16 QUESTIONS

REMEMBER Subtracting a **POSITIVE**: move left.**1** $7 - 11$ **2** $-4 - 8$ **3** $2 - 9$ **4** $-6 - 1$ **5** $0 - 5$ **6** $-10 - 10$

Questions 7–16: evaluate.

7 $10 - (-3)$ **8** $-5 - (-8)$ **9** $-3 - 8$ **10** $-6 - 4$ **11** $2 - (-9)$ **12** $-11 - (-4)$ **13** $0 - 7$ **14** $14 - (-6)$ **15** $-15 - 3$ **16** $-20 - (-5)$

YOU DO

On your own: **Easy** Set B. Subtracting a negative: change to adding.

6 QUESTIONS

1 $3 - (-4)$

4 $8 - (-2)$

2 $-5 - (-9)$

5 $-12 - (-3)$

3 $-5 - (-5)$

6 $0 - (-7)$

Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $-3 - 4 - 5$

Your teacher works this out. Copy every step.

b $8 - 6 - (-4)$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate each expression.

6 QUESTIONS

REMEMBER Subtracting a **NEGATIVE** = adding the positive.

1 $-3 - 4 - 5$

4 $3 - 8 + (-6)$

2 $-8 + 5 - 4$

5 $-2 - (-7) - 10$

3 $8 - 6 - (-4)$

6 $15 - (-5) + (-20)$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** The temperature rose from -9°C to 6°C . By how much?

Your teacher works this out. Copy every step.

- b** Find the missing number: $-2 - \square = 6$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Subtraction in context.

4 QUESTIONS

REMEMBER $-(-)$ becomes $+$.

- 1** The minimum temperature was -5°C and the maximum was 11°C . By how much did it rise?

- 3** A submarine at -120 m rises 45 m. How deep is it now?

- 2** Find the difference between -18 m and 7 m.

- 4** What is 4 less than -9 ?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-3 - 6$

Level 2

$4 - (-9)$

Level 3

The temperature went from -7°C to 12°C . How much did it rise?

 Extension · Stretch yourself Worked example

Find the missing number: $6 - \square = 10$

1. Subtracting a negative adds
2. $6 + 4 = 10$, so we subtracted -4
3. Check: $6 - (-4) = 10$ ✓

Answer: -4

YOU DO

On your own: read the worked example, then try these.

E1 $-3 - \square = 5$

E2 $\square - (-8) = -2$

Lesson summary

Missed the lesson? Start here.

Remember

1. Subtracting a **positive**: move left.
3. $-(-)$ becomes $+$.

2. Subtracting a **negative** = adding the positive.
4. Difference between two temperatures: higher – lower.

 Level 1

$5 - 9$

1. Start at 5, move 9 left

Answer: -4

 Level 2

$-2 - (-6)$

1. $= -2 + 6$

Answer: 4

 Level 3

$-4 - 3 - (-8)$

1. $-4 - 3 = -7$

2. $-7 + 8$

Answer: 1

 Extension

Rise from -8°C to 5°C

1. $5 - (-8) = 13$

Answer: 13°C

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.06

Multiplying integers

Date _____

Learning intentions

We are learning to:

- multiply positive and negative integers

Success criteria

I can:

- ☐ know the sign rules for multiplying
- ☐ multiply two integers
- ☐ multiply three integers by working in pairs

Key terms Write the definition as your teacher explains it.

Product

Sign rule

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate each product.

a $6 \times (-3)$ and -2×5

Your teacher works this out. Copy every step.

b $-8 \times (-6)$ and $-6 \times (-4) \times (-1)$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each product.

16 QUESTIONS

REMEMBER $++ = +$

1 $6 \times (-3)$

2 -2×5

3 -11×4

4 $7 \times (-9)$

5 -3×8

6 $10 \times (-10)$

Questions 7–16: evaluate each product.

7 $-3 \times (-5)$

8 -7×2

9 $8 \times (-3)$

10 $-9 \times (-2)$

11 $5 \times (-6)$

12 -4×7

13 $-10 \times (-10)$

14 $12 \times (-2)$

15 -8×4

16 $-6 \times (-6)$

YOU DO

On your own: **Easy** Set B. Negative × negative.

6 QUESTIONS

1 $-8 \times (-6)$

4 $-12 \times (-2)$

2 $-5 \times (-5)$

5 $-1 \times (-20)$

3 $-7 \times (-3)$

6 $-9 \times (-4)$

Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $-2 \times 3 \times 7$

Your teacher works this out. Copy every step.

b $-4 \times 2 \times (-5)$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate each product.

6 QUESTIONS

REMEMBER $- \times - = +$

1 $-6 \times (-4) \times (-1)$

4 $(-3)^2$

2 $-2 \times 3 \times 7$

5 $-5 \times (-2) \times 10$

3 $-4 \times 2 \times (-5)$

6 $(-2) \times (-2) \times (-2)$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the missing number, or explain.

a Find the missing number: $\square \times (-7) = 56$

Your teacher works this out. Copy every step.

b Is $(-1)^{100}$ positive or negative? Explain.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER $+ \times - = -$

1 A diver descends 4 m every minute. Where is she after 7 minutes?

3 Is the product of 5 negative numbers positive or negative? Explain.

2 Find the missing number: $-6 \times \square = 42$

4 The temperature drops 3°C each hour for 6 hours. What is the total change?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

-7×6

Level 2

$-8 \times (-9)$

Level 3

$-3 \times (-4) \times (-5)$

 Extension · Stretch yourself Worked example

Evaluate $(-1) \times (-1) \times (-1) \times \dots$ with 15 factors of -1 .

1. Each pair of -1 s makes $+1$
2. 15 factors = 7 pairs and one left over

Answer: -1

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $(-2)^5$

E2 Find two integers with a product of -24 and a sum of 5 .

Lesson summary

Missed the lesson? Start here.

Remember 1. $++ = +$
3. $+- = -$

2. $-- = +$
4. Odd number of negatives $\rightarrow$ negative answer.

 Level 1

$$4 \times (-7)$$

1. Different signs $\rightarrow$ negative

Answer: -28

 Level 2

$$-9 \times (-3)$$

1. Same signs $\rightarrow$ positive

Answer: 27

 Level 3

$$-2 \times (-5) \times (-3)$$

1. $-2 \times (-5) = 10$
2. $10 \times (-3)$

Answer: -30

 Extension

$$(-3)^3$$

1. Three negatives $\rightarrow$ negative
2. 27

Answer: -27

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.07

Dividing integers

Date _____

Learning intentions

We are learning to:

- divide positive and negative integers

Success criteria

I can:

- ☐ use the same sign rules for dividing as for multiplying
- ☐ divide two integers
- ☐ work left to right with two divisions

Key terms Write the definition as your teacher explains it.

Quotient

Sign rule

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate each quotient.

a $-12 \div (-4)$ and $15 \div (-5)$

Your teacher works this out. Copy every step.

b $24 \div (-6) \div (-2)$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each quotient.

16 QUESTIONS

REMEMBER Same signs $\rightarrow$ positive.

1 $-12 \div (-4)$

2 $15 \div (-5)$

3 $-6 \div (-6)$

4 $16 \div (-4)$

5 $-24 \div 12$

6 $-1000 \div 10$

Questions 7–16: evaluate each quotient.

7 $-36 \div 9$

8 $-24 \div (-6)$

9 $45 \div (-5)$

10 $-40 \div 8$

11 $-18 \div (-3)$

12 $-56 \div 7$

13 $30 \div (-6)$

14 $-27 \div (-9)$

15 $63 \div (-9)$

16 $-72 \div (-8)$

YOU DO

On your own: **Easy** Set B. Evaluate each quotient.

6 QUESTIONS

1 $-45 \div 9$

4 $-100 \div (-25)$

2 $-56 \div (-8)$

5 $0 \div (-3)$

3 $63 \div (-7)$

6 $-81 \div 9$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Work from left to right.

a $-36 \div 3 \div 4$

Your teacher works this out. Copy every step.

b $-64 \div (-4) \div 2$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Work from left to right.

6 QUESTIONS

REMEMBER Different signs → negative.

1 $24 \div (-6) \div (-2)$

4 $-60 \div (-5) \div (-3)$

2 $-36 \div 3 \div 4$

5 $100 \div (-10) \div (-5)$

3 $-64 \div (-4) \div 2$

6 $-48 \div 2 \div (-3)$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve.

- a** A debt of \$96 is shared by 8 people. Write each share as an integer.

Your teacher works this out. Copy every step.

- b** Find the missing number: $\square \div (-6) = -9$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Division in context.

4 QUESTIONS

REMEMBER 0 divided by any number is 0.

- 1** A \$-84 debt is shared equally by 4 people. What is each share?

- 3** Find the missing number: $\square \div (-5) = 7$

- 2** The temperature fell 18°C over 6 hours. What was the change each hour?

- 4** Find the missing number: $-72 \div \square = 9$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-36 \div 9$

Level 2

$-42 \div (-6)$

Level 3

$-80 \div (-4) \div (-5)$

 Extension · Stretch yourself Worked exampleEvaluate $-6 \times 8 \div (-12)$

1. Work left to right
2. $-6 \times 8 = -48$
3. $-48 \div (-12) = 4$

Answer: 4

YOU DO

On your own: read the worked example, then try these.

E1 $-90 \div (-3) \times (-2)$

E2 $-5 \times (-12) \div (-4) \div 3$

Lesson summary

Missed the lesson? Start here.

Remember

1. Same signs $\rightarrow$ positive.
2. Different signs $\rightarrow$ negative.
3. 0 divided by any number is 0.
4. Two divisions: work left to right.

 Level 1

$-20 \div 4$

1. Different signs

Answer: -5

 Level 2

$-35 \div (-7)$

1. Same signs

Answer: 5

 Level 3

$-40 \div (-2) \div (-5)$

1. $-40 \div (-2) = 20$
2. $20 \div (-5)$

Answer: -4

 Extension

$\square \div (-3) = -6$

1. $-6 \times (-3)$

Answer: 18

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LESSON
1.08

Order of operations with integers

Date _____

Learning intentions

We are learning to:

- use the order of operations with positive and negative numbers

Success criteria

I can:

- ☐ do brackets first, then $\times$ and $\div$, then $+$ and $-$
- ☐ work left to right for operations of the same kind
- ☐ evaluate an expression with a fraction bar

Key terms Write the definition as your teacher explains it.

Order of operations

Fraction bar

Notes

Level 1 · Getting started

Easy · 22 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate. Use the order of operations.

a $-6 + (-3) \times 2$

Your teacher works this out. Copy every step.

b $(9 - 17) \div (-4 + 6)$, written as a fraction

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each expression.

16 QUESTIONS

REMEMBER Brackets first.

1 $4 - 3 \times 6$

2 $-12 \div 4 - 10$

3 $30 \div (-6) + 5$

4 $-2 + 5 \times 3$

5 $10 - (-2) \times 4$

6 $-8 \div 2 + 1$

Questions 7–16: evaluate. $\times$ and $\div$ before $+$ and $-$.

7 $-3 + 4 \times 2$

8 $5 - 3 \times (-4)$

9 $-10 \div 2 + (-3)$

10 $8 + (-2) \times 3$

11 $-6 \times 2 + 5$

12 $12 \div (-4) - 1$

13 $-2 + (-3) \times 4$

14 $9 - 6 \div 3$

15 $4 \times (-3) - 2$

16 $7 + 8 \div (-2)$

YOU DO On your own: **Easy** Set B. Evaluate each expression.

6 QUESTIONS

1 $-3 \times 5 + 6 \div (-2)$

4 $20 \div (-5 - 5)$

2 $-2 \times 7 + 15 \div (-3)$

5 $-3 \times (2 - 7)$

3 $(-4 + 1) \times 5$

6 $(-6 - 2) \div (-4)$

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Evaluate. Brackets first.

a $-3 \times 5 + 6 \div (-2)$

Your teacher works this out. Copy every step.

b $-12 + (-3) \times (-7) - 2$

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Evaluate each expression.

6 QUESTIONS

REMEMBER $\times$ and $\div$ from left to right.

1 $-12 + (-3) \times (-7) - 2$

4 $-20 \div (-2 \times 5)$

2 $-6 + (-6) \div (-2) \times 2$

5 $(-3)^2 - 10$

3 $5 - 3 \times (-2) + 1$

6 $4 \times (-3) - (-2) \times 5$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Show every step.

a $\frac{7 + (-12) + 6}{2 - 5}$

b $-6 + (-6) \div (-2) \times 2$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Fraction bars: work out the top and bottom first.

4 QUESTIONS

REMEMBER + and – from left to right.

1 $\frac{9 - 17}{-4 + 6}$

3 $\frac{-3 \times 8}{-6}$

2 $\frac{7 + (-12) + 6}{2 - 5}$

4 $\frac{-10 - 14}{2 \times (-3)}$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-5 + 2 \times 3$

Level 2

$-3 \times (4 - 9)$

Level 3

$\frac{-4 \times 6}{3 - 5}$

Extension · Stretch yourself

Worked example

Insert brackets to make this true: $-2 + 6 \times -3 = -12$

1. Try $(-2 + 6) \times -3$
2. $= 4 \times (-3) = -12$ ✓

Answer: $(-2 + 6) \times -3$

YOU DO

On your own: read the worked example, then try these.

- E1** Insert brackets: $8 - 12 \div -4 + 2 = 2$

- E2** Write an expression using -3 , 4 and 5 that equals -17 .

Lesson summary

Missed the lesson? Start here.

Order of operations

1. Brackets first.
2. $\times$ and $\div$ from left to right.
3. $+$ and $-$ from left to right.
4. Fraction bar: work out top and bottom first.

Level 1

$-5 + 4 \times (-2)$

1. $4 \times (-2) = -8$
2. $-5 + (-8)$

Answer: -13

Level 2

$(-3 - 5) \times 2$

1. $-3 - 5 = -8$
2. -8×2

Answer: -16

Level 3

$\frac{-18}{2 - 5}$

1. $2 - 5 = -3$
2. $-18 \div (-3)$

Answer: 6

Extension

$(-2 + 6) \times -3$

1. 4×-3

Answer: -12

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.09

Integers on a calculator

Date _____

Learning intentions

We are learning to:

- use a calculator to work with integers

Success criteria

I can:

- ☐ enter a negative number using the $(-)$ key
- ☐ use brackets on a calculator
- ☐ check that an answer is reasonable

Key terms Write the definition as your teacher explains it.

$(-)$ key

Estimate

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use your calculator to evaluate.

a Use your calculator: $28 - 37$ and $-17 + 11$

b $[5 + (-20)] \div (-3)$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Use your calculator.

14 QUESTIONS

REMEMBER Use the $(-)$ key for negative numbers.

1 $28 - 37$

2 $-17 + 11$

3 $-45 - 38$

4 -16×3

5 $-144 \div (-12)$

6 $-250 + 175$

Questions 7–14: use your calculator.

7 $-127 + 58$

8 $-93 - 47$

9 -36×25

10 $-1260 \div (-35)$

11 $482 - 715$

12 $-19 \times (-21)$

13 $-2048 \div 64$

14 $-350 + 1275$

YOU DO

On your own: **Easy** Set B. Use your calculator.

6 QUESTIONS

1 $18 - 15 \times 2$

4 $-7 \times (-8) - 60$

2 $-16 \times 3 + (-4)$

5 $-96 \div 8 + 5$

3 $100 - 120 \div 4 + 17$

6 $(-13)^2$

 Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use your calculator. Enter the brackets.

a $18 - 15 \times 2$ on a calculator

Your teacher works this out. Copy every step.

b $100 - 120 \div 4 + 17$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Use brackets on your calculator.

6 QUESTIONS

REMEMBER Calculators follow the order of operations.

1 $[5 + (-20)] \div (-3)$

[illegible]

④ $(-8 + 3) \times (-6 - 1)$

[illegible]

2 $(-14 - 22) \div (-4)$

5 $(-55 + 19) \div 12$

[illegible]

$$3 \quad -3 \times (17 - 29)$$

[illegible]

6 $-(-7 - 11) \times 2$

[illegible]

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use your calculator.

a $-16 \times 3 + (-4)$

b
$$\frac{-84 - 16}{-25}$$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Estimate first, then check with a calculator.

4 QUESTIONS

REMEMBER Use brackets for fraction bars.

1 Estimate -49×21 , then calculate.

3 Jo typed -6^2 and got -36 . Why? How do you get 36 ?

2 Estimate $-612 \div 29.8$ (1 d.p.), then calculate.

4 Find the error: $20 - 4 \times 3 = 48$ on a calculator?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-84 + 39$

Level 2

$-15 \times (-7) - 100$

Level 3

$(-35 - 45) \div (-16)$

Extension · Stretch yourself**Worked example**

Evaluate $\frac{-48 + 12}{-3 \times (-4)}$ on a calculator.

1. Put brackets around the top and the bottom
2. $(-48 + 12) \div (-3 \times (-4))$
3. $= -36 \div 12$

Answer: -3

YOU DO

On your own: read the worked example, then try these.

E1 $\frac{-125 - 25}{6 \times (-5)}$

E2 $\frac{(-9)^2 - 1}{-16}$

Lesson summary

Missed the lesson? Start here.

Remember

1. Use the $(-)$ key for negative numbers.
2. Calculators follow the order of operations.
3. Use brackets for fraction bars.
4. Estimate to check your answer.

Level 1

$$-23 + 9$$

1. $(-) 23 + 9 =$

Answer: -14

Level 2

$$18 - 15 \times 2$$

1. $\times$ first

Answer: -12

Level 3

$$(-36 \div 6) \div 5$$

1. Brackets on top

Answer: -6

Extension

$$(-3)^2 \text{ vs } -3^2$$

1. $(-3)^2 = 9$, $-3^2 = -9$

Answer: 9 and -9

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.10

Integer problems

Date _____

Learning intentions

We are learning to:

- solve real problems using integers

Success criteria

I can:

- ☐ decide which operation a problem needs
- ☐ write an integer calculation for a word problem
- ☐ check the answer makes sense

Key terms Write the definition as your teacher explains it.

Rise

Balance

Deposit

Notes

Level 1 · Getting started

Easy · 8 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Write a number sentence, then answer.

- a** The minimum overnight temperature in Goulburn was -5°C . During the day it rose to 11°C . By how much did it rise?

Your teacher works this out. Copy every step.

- b** A bank account has \$120. Three withdrawals of \$55 are made. What is the balance?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Temperature problems.

4 QUESTIONS

REMEMBER Read carefully and underline the numbers.

- 1** It was -3°C at 6 am and 14°C at noon. What was the rise?

- 3** The temperature fell 2°C each hour from 5°C . What is it after 4 hours?

- 2** It was 8°C and fell 12°C . What is the new temperature?

- 4** Find the average of -4°C , 2°C , -1°C and -5°C .

YOU DO On your own: **Easy** Set B. Heights and depths.

4 QUESTIONS

- 1 A diver at -18 m rises 7 m. How deep is she?

- 2 A lift goes from level 3 to level -2 . How many floors did it travel?

- 3 A plane at 900 m descends 350 m. What is its height now?

- 4 A cave is 35 m below sea level and a hill is 120 m above. Find the difference.

Level 2 · Building up

Medium · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Write a number sentence, then answer.

- a A lift at level 4 goes down 7 floors. Which level is it on?

Your teacher works this out. Copy every step.

- b Balance $-\$35$, deposit $\$120$. New balance?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Money problems.

4 QUESTIONS

REMEMBER Rise, gain, deposit $\rightarrow$ add. Fall, loss, withdraw $\rightarrow$ subtract.

- 1 Balance $\$40$, withdraw $\$65$. New balance?

- 3 Sam owes $\$12$ to each of 5 friends. Write this as an integer and find the total.

- 2 Balance $-\$30$, deposit $\$75$. New balance?

- 4 A shop made $-\$250$ on Monday and $\$400$ on Tuesday. Total?

Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write a number sentence, then answer.

- a** Harrison saves \$80 a week and pays \$210 a month for a car loan. How much after 6 months (26 weeks)?

Your teacher works this out. Copy every step.

- b** A diver at -42 m rises 6 m every minute. Where is she after 5 minutes?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Multi-step problems.

2 QUESTIONS

REMEMBER Difference: larger – smaller.

- 1** Harrison saves \$80 a week. Each month \$210 is taken out for a car loan. How much is in the account after 6 months (26 weeks)?

- 2** A game scores +5 for a win and -3 for a loss. Mia wins 7 and loses 9. What is her score?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

It was -6°C and rose 10°C . What is the temperature now?

Level 2

A diver goes from -30 m to -12 m. How far did she rise?

Level 3

A score starts at 0. Gain 4, lose 7, lose 7, gain 3. Final score?

Extension · Stretch yourself**Worked example**

A submarine at -240 m rises 15 m per minute.
How long until it is at -60 m?

1. Distance to rise: $-60 - (-240) = 180$ m
2. Time: $180 \div 15 = 12$

Answer: 12 minutes

YOU DO

On your own: read the worked example,
then try these.

- E1** A freezer is at 4°C and cools by 3°C every 10 minutes. How long until it is -17°C ?

- E2** Write your own integer problem with an answer of -8 .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Read carefully and underline the numbers.
 2. Rise, gain, deposit $\rightarrow$ add. Fall, loss, withdraw $\rightarrow$ subtract.
 3. Difference: larger $-$ smaller.
 4. Check the answer makes sense.

Level 1

From -5°C to 11°C

1. $11 - (-5)$

Answer: 16°C rise

Level 2

$\$20$ balance, withdraw
 $\$50$

1. $20 - 50$

Answer: $-\$30$

Level 3

6 months: $+\$80 \times 26$, $-\$210 \times 6$

1. $2080 - 1260$

Answer: $\$820$

Extension

Rise 15 m/min from
 -240 m to -60 m

1. $180 \div 15$

Answer: 12 minutes

This space is left blank on purpose. It is the back of your exit ticket.